

# Signed sub-optimality bound for sample-path AIS

## Extended Theorem 3

**Lemma 1** (Policy evaluation). *Let  $\hat{\pi} = (\hat{\pi}_1, \dots, \hat{\pi}_T)$ ,  $\hat{\pi}_t : \mathcal{Z} \rightarrow \mathcal{A}$ , be any AIS policy and  $\pi_t = \pi_t \circ \sigma$ ,  $\pi_t : \mathcal{H} \rightarrow \mathcal{A}$  its lifted version with  $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$  for all  $t \in \{1, \dots, T\}$ . Define*

$$\alpha_t^{\hat{\pi}}(h_t) = V_t^{\pi}(h_t) - \hat{V}^{\hat{\pi}}(\sigma_t(h_t)), \quad (1)$$

$$\beta_t^{\hat{\pi}}(h_t, a_t) = Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t), \quad (2)$$

and

$$\Delta_t^{\hat{\pi}}(h_t, a_t) = \left[ c_t(h_t, a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) \right] \quad (3)$$

$$- \left[ \hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1} | h_t, a_t) \right]. \quad (4)$$

Then, for all  $h_t \in \mathcal{H}_t$  and  $a_t \in \mathcal{A}$ , the following holds

$$\beta_t^{\hat{\pi}}(h_t, a_t) = \Delta_t^{\hat{\pi}}(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (5)$$

$$\alpha_t^{\hat{\pi}}(h_t) = \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))). \quad (6)$$

where  $\alpha_{T+1}^{\hat{\pi}} \equiv 0$ .

*Proof.* We first establish the recursion (5) and then the identity (6).

*Step 1 (the  $\beta^{\hat{\pi}}$ -recursion).* Fix  $t \in \{1, \dots, T\}$ ,  $h_t \in \mathcal{H}_t$ , and  $a_t \in \mathcal{A}$ . Adding and subtracting  $\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t)$ , and using  $\hat{P}_t(\cdot | \sigma_t(h_t), a_t) = \hat{P}_t(\cdot | h_t, a_t)$ , we get

$$\begin{aligned} \beta_t^{\hat{\pi}}(h_t, a_t) &= Q_t^{\pi}(h_t, a_t) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), a_t) \\ &= [c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] \\ &\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t(dz_{t+1} | \sigma_t(h_t), a_t) \\ &= \underbrace{[c_t(h_t, a_t) - \hat{c}_t(\sigma_t(h_t), a_t)] + \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) \hat{P}_t^{\sigma}(dz_{t+1} | h_t, a_t)}_{\Delta_t^{\hat{\pi}}(h_t, a_t)} \\ &\quad + \int V_{t+1}^{\pi}(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t), \end{aligned}$$

where the first bracket is  $\Delta_t^{\hat{\pi}}(h_t, a_t)$  by (3). For the remaining terms, we have

$$\int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^{\sigma}(dz_{t+1} | h_t, a_t) = \int \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) P_{Y,t}(dy_{t+1} | h_t, a_t).$$

Thus the last term can be written as

$$\begin{aligned} & \int V_{t+1}^\pi(h_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) \\ & - \int \hat{V}_{t+1}^{\hat{\pi}}(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t) = \int \left[ V_{t+1}^\pi(h_{t+1}) - \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) \right] P_{Y,t}(dy_{t+1} | h_t, a_t) \\ & = \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t) \end{aligned}$$

so the two integrals against  $P_{Y,t}$  combine into

$$\int \left[ V_{t+1}^\pi(h_{t+1}) - \hat{V}_{t+1}^{\hat{\pi}}(\sigma_{t+1}(h_{t+1})) \right] P_{Y,t}(dy_{t+1} | h_t, a_t) = \int \alpha_{t+1}^{\hat{\pi}}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t),$$

This establishes (5). At  $t = T$ , we have  $\alpha_{T+1}^{\hat{\pi}} \equiv 0$  and  $\Delta_T^{\hat{\pi}}(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$ , so (5) reads  $\beta_T^{\hat{\pi}} = \Delta_T^{\hat{\pi}}$ .

*Step 2 (the  $\alpha^{\hat{\pi}}$ -identity).* Since  $\hat{\pi}_t$  is deterministic and  $\pi_t(h_t) = \hat{\pi}_t(\sigma_t(h_t))$ , the value functions are the  $Q$ -functions evaluated at the prescribed actions:

$$V_t^\pi(h_t) = Q_t^\pi(h_t, \hat{\pi}_t(\sigma_t(h_t))), \quad \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) = \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))).$$

Then, we can write

$$\alpha_t^{\hat{\pi}^*}(h_t) = V_t^\pi(h_t) - \hat{V}_t^{\hat{\pi}}(\sigma_t(h_t)) \tag{7}$$

$$= Q_t^\pi(h_t, \hat{\pi}_t(\sigma_t(h_t))) - \hat{Q}_t^{\hat{\pi}}(\sigma_t(h_t), \hat{\pi}_t(\sigma_t(h_t))) \tag{8}$$

$$= \beta_t^{\hat{\pi}}(h_t, \hat{\pi}_t(\sigma_t(h_t))) \tag{9}$$

□

**Lemma 2** (Optimality). *Define*

$$\alpha_t^*(h_t) = V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \tag{10}$$

$$\beta_t^*(h_t, a_t) = Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t), \tag{11}$$

and

$$\Delta_t(h_t, a_t) = \left[ c_t(h_t, a_t) + \int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t) \right] \tag{12}$$

$$- \left[ \hat{c}_t(\sigma_t(h_t), a_t) + \int \hat{V}_{t+1}^*(z_{t+1}) \hat{P}_t^\sigma(dz_{t+1} | h_t, a_t) \right]. \tag{13}$$

Then, the following holds

$$\beta_t^*(h_t, a_t) = \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \tag{14}$$

$$\beta_t^*(h_t, \pi_t^*(h_t)) \leq \alpha_t^*(h_t) \leq \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \tag{15}$$

where  $\alpha_{T+1}^* \equiv 0$ .

*Proof. Step 1 (the  $\beta^*$ -recursion).* [Complete Step 1, similar to Lemma 1.1]

The derivation is identical to Step 1 in the proof of Lemma 1 with  $(Q_t^\pi, \hat{Q}_t^\pi, V_{t+1}^\pi, \hat{V}_{t+1}^\pi)$  replaced by  $(Q_t^*, \hat{Q}_t^*, V_{t+1}^*, \hat{V}_{t+1}^*)$ , add and subtract  $\int \hat{V}_{t+1}^*(z_{t+1}) P_t^\sigma(dz_{t+1} | h_t, a_t)$ , and identify the cost-and-kernel-mismatch bracket as  $\Delta_t(h_t, a_t)$  from (12).

*Step 2 (the  $\alpha^*$ -inequalities).* The two inequalities in (15) follow from the two optimality properties, used one at a time.

*Upper bound.* By optimality of  $\hat{\pi}_t^*$  on the AIS side,  $\hat{V}_t^*(\sigma_t(h_t)) = \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t)))$ , while on the true side  $V_t^*(h_t) = \inf_{a \in \mathcal{A}} Q_t^*(h_t, a) \leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t)))$ . Therefore,

$$\begin{aligned} \alpha_t^*(h_t) &= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ &\leq Q_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) - \hat{Q}_t^*(\sigma_t(h_t), \hat{\pi}_t^*(\sigma_t(h_t))) = \beta_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \end{aligned}$$

*Lower bound.* By optimality of  $\pi_t^*$  on the true side,  $V_t^*(h_t) = Q_t^*(h_t, \pi_t^*(h_t))$ , while on the AIS side  $\hat{V}_t^*(\sigma_t(h_t)) = \inf_{a \in \mathcal{A}} \hat{Q}_t^*(\sigma_t(h_t), a) \leq \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t))$ . Therefore,

$$\begin{aligned} \alpha_t^*(h_t) &= V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)) \\ &\geq Q_t^*(h_t, \pi_t^*(h_t)) - \hat{Q}_t^*(\sigma_t(h_t), \pi_t^*(h_t)) = \beta_t^*(h_t, \pi_t^*(h_t)). \end{aligned}$$

Combining the two gives (15).  $\square$

**Theorem 3.** Suppose that for each  $t \in \{1, \dots, T\}$ , there exist measurable functions  $\alpha_t^{\hat{\pi}^*}: \mathcal{H}_t \rightarrow \mathbb{R}$ ,  $\alpha_t^*: \mathcal{H}_t \rightarrow \mathbb{R}$ ,  $\beta_t^{\hat{\pi}^*}: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ , and  $\beta_t^*: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$  such that

$$\beta_T^{\hat{\pi}^*}(h_T, a_T) \geq \Delta_T(h_T, a_T) \geq \beta_T^*(h_T, a_T) \quad (16)$$

and for  $t < T$ :

$$\beta_t^{\hat{\pi}^*}(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (17)$$

$$\beta_t^*(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (18)$$

and for all  $t \in \{1, \dots, T\}$ :

$$\alpha_t^{\hat{\pi}^*}(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (19)$$

$$\alpha_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)), \quad (20)$$

where  $\Delta_t$  is as defined in (12) (in particular, at terminal time,  $\Delta_T(h_T, a_T) = c_T(h_T, a_T) - \hat{c}_T(\sigma_T(h_T), a_T)$ ), and  $\pi_t^*$  and  $\hat{\pi}_t^*$  are as in Lemma 2. Then, for all  $t \in \{1, \dots, T\}$  and all  $h_t \in \mathcal{H}_t$ , the AIS-based policy  $\pi_t(h_t) = \hat{\pi}_t^*(\sigma_t(h_t))$  satisfies

$$V_t^\pi(h_t) - V_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t). \quad (21)$$

Symmetrically, suppose that for each  $t \in \{1, \dots, T\}$  there exist measurable functions  $\underline{\alpha}_t^{\hat{\pi}^*}: \mathcal{H}_t \rightarrow \mathbb{R}$ ,  $\underline{\alpha}_t^*: \mathcal{H}_t \rightarrow \mathbb{R}$ ,  $\underline{\beta}_t^{\hat{\pi}^*}: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$ , and  $\underline{\beta}_t^*: \mathcal{H}_t \times \mathcal{A} \rightarrow \mathbb{R}$  such that

$$\underline{\beta}_T^{\hat{\pi}^*}(h_T, a_T) \leq \Delta_T(h_T, a_T) \leq \underline{\beta}_T^*(h_T, a_T) \quad (22)$$

and for  $t < T$ :

$$\underline{\beta}_t^{\hat{\pi}^*}(h_t, a_t) \leq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (23)$$

$$\underline{\beta}_t^*(h_t, a_t) \geq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} | h_t, a_t), \quad (24)$$

and for all  $t \in \{1, \dots, T\}$ :

$$\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \underline{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (25)$$

$$\underline{\alpha}_t^*(h_t) \geq \underline{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))). \quad (26)$$

Then, for all  $t \in \{1, \dots, T\}$  and all  $h_t \in \mathcal{H}_t$ ,

$$V_t^\pi(h_t) - V_t^*(h_t) \geq \underline{\alpha}_t^{\hat{\pi}^*}(h_t) - \underline{\alpha}_t^*(h_t). \quad (27)$$

*Proof.* [Check proof]

Throughout the proof, let  $\tilde{\alpha}_t^{\hat{\pi}^*}$ ,  $\tilde{\beta}_t^{\hat{\pi}^*}$ ,  $\tilde{\alpha}_t^*$ , and  $\tilde{\beta}_t^*$  denote the *exact* signed error functions of Lemmas 1 and 2:

$$\tilde{\alpha}_t^{\hat{\pi}^*}(h_t) = V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \quad \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t) = Q_t^\pi(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 1 specialized to  $\hat{\pi} = \hat{\pi}^*$ , and

$$\tilde{\alpha}_t^*(h_t) = V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t)), \quad \tilde{\beta}_t^*(h_t, a_t) = Q_t^*(h_t, a_t) - \hat{Q}_t^*(\sigma_t(h_t), a_t),$$

i.e., the functions of Lemma 2. Regarding the specialization  $\hat{\pi} = \hat{\pi}^*$  in Lemma 1: since  $\hat{\pi}_t^*(z_t)$  is a minimizer of  $\hat{Q}_t^*(z_t, \cdot)$ , evaluating the AIS dynamic program along  $\hat{\pi}^*$  shows that the policy-evaluation value functions of  $\hat{\pi}^*$  coincide with the optimal ones,  $\hat{V}_t^{\hat{\pi}^*} = \hat{V}_t^*$  and  $\hat{Q}_t^{\hat{\pi}^*} = \hat{Q}_t^*$  for all  $t$ ; consequently, the integrand  $\hat{V}_{t+1}^{\hat{\pi}^*}$  in (3) equals  $\hat{V}_{t+1}^*$  and the mismatch terms coincide,  $\Delta_t^{\hat{\pi}^*} = \Delta_t$ , the same  $\Delta_t$  as in Lemma 2 and in the statement of the theorem. Thus Lemmas 1 and 2 give, for all  $t$ ,  $h_t$ , and  $a_t$  (with  $\tilde{\alpha}_{T+1}^{\hat{\pi}^*} \equiv 0$  and  $\tilde{\alpha}_{T+1}^* \equiv 0$ ):

$$\begin{aligned} \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\ \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) &= \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \end{aligned} \quad (28)$$

$$\begin{aligned} \tilde{\beta}_t^*(h_t, a_t) &= \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_t, a_t, y_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t), \\ \tilde{\beta}_t^*(h_t, \pi_t^*(h_t)) &\leq \tilde{\alpha}_t^*(h_t). \end{aligned} \quad (29)$$

*Step 1 (the candidate functions compared to the exact errors).* We show by backward induction that for all  $t \in \{1, \dots, T\}$  and all  $h_t \in \mathcal{H}_t$ ,

$$\alpha_t^{\hat{\pi}^*}(h_t) \geq \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) \quad \text{and} \quad \alpha_t^*(h_t) \leq \tilde{\alpha}_t^*(h_t). \quad (30)$$

*Base case* ( $t = T$ ). Since  $\tilde{\alpha}_{T+1}^{\hat{\pi}^*} \equiv 0$  and  $\tilde{\alpha}_{T+1}^* \equiv 0$ , the recursions in (28) and (29) give  $\tilde{\beta}_T^{\hat{\pi}^*} = \tilde{\beta}_T^* = \Delta_T$ . Hence, by (16),  $\beta_T^{\hat{\pi}^*} \geq \tilde{\beta}_T^{\hat{\pi}^*}$  and  $\beta_T^* \leq \tilde{\beta}_T^*$ . Therefore, by (19) and the identity in (28),

$$\alpha_T^{\hat{\pi}^*}(h_T) \geq \beta_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) = \tilde{\alpha}_T^{\hat{\pi}^*}(h_T),$$

and, by (20) and the sandwich inequality in (29),

$$\alpha_T^*(h_T) \leq \beta_T^*(h_T, \pi_T^*(h_T)) \leq \tilde{\beta}_T^*(h_T, \pi_T^*(h_T)) \leq \tilde{\alpha}_T^*(h_T).$$

*Induction step* ( $t < T$ ). Suppose (30) holds at  $t + 1$ . Then, by (17), the induction hypothesis, and the recursion in (28),

$$\begin{aligned} \beta_t^{\hat{\pi}^*}(h_t, a_t) &\geq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\geq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t), \end{aligned}$$

and then, exactly as in the base case,  $\alpha_t^{\hat{\pi}^*}(h_t) \geq \beta_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t)$ . Symmetrically, by (18), the induction hypothesis, and the recursion in (29),

$$\begin{aligned}\beta_t^*(h_t, a_t) &\leq \Delta_t(h_t, a_t) + \int \alpha_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\leq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^*(h_t, a_t),\end{aligned}$$

and then  $\alpha_t^*(h_t) \leq \beta_t^*(h_t, \pi_t^*(h_t)) \leq \tilde{\beta}_t^*(h_t, \pi_t^*(h_t)) \leq \tilde{\alpha}_t^*(h_t)$ . This establishes (30) for all  $t$ .

*Step 2 (bridge identity).* Adding and subtracting  $\hat{V}_t^*(\sigma_t(h_t))$ ,

$$V_t^\pi(h_t) - V_t^*(h_t) = [V_t^\pi(h_t) - \hat{V}_t^*(\sigma_t(h_t))] - [V_t^*(h_t) - \hat{V}_t^*(\sigma_t(h_t))] = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t).$$

Combining Step 2 with (30),

$$V_t^\pi(h_t) - V_t^*(h_t) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t) \leq \alpha_t^{\hat{\pi}^*}(h_t) - \alpha_t^*(h_t),$$

which is (21).

*Step 3 (the lower bound).* The argument mirrors Steps 1–2 with every inequality reversed; the only structural change is that the  $\star$ -side now invokes the *upper* side of the sandwich in Lemma 2, namely

$$\tilde{\alpha}_t^*(h_t) \leq \tilde{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))), \quad (31)$$

in place of (29). We show by backward induction that for all  $t$  and all  $h_t \in \mathcal{H}_t$ ,

$$\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) \quad \text{and} \quad \underline{\alpha}_t^*(h_t) \geq \tilde{\alpha}_t^*(h_t). \quad (32)$$

*Base case ( $t = T$ ).* As before  $\tilde{\beta}_T^{\hat{\pi}^*} = \tilde{\beta}_T^* = \Delta_T$ , so by (22),  $\underline{\beta}_T^{\hat{\pi}^*} \leq \tilde{\beta}_T^{\hat{\pi}^*}$  and  $\underline{\beta}_T^* \geq \tilde{\beta}_T^*$ . Therefore, by (25) and the identity in (28),

$$\underline{\alpha}_T^{\hat{\pi}^*}(h_T) \leq \underline{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \leq \tilde{\beta}_T^{\hat{\pi}^*}(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) = \tilde{\alpha}_T^{\hat{\pi}^*}(h_T),$$

and, by (26) and the upper sandwich inequality (31),

$$\underline{\alpha}_T^*(h_T) \geq \underline{\beta}_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\beta}_T^*(h_T, \hat{\pi}_T^*(\sigma_T(h_T))) \geq \tilde{\alpha}_T^*(h_T).$$

*Induction step ( $t < T$ ).* Suppose (32) holds at  $t + 1$ . Then, by (23), the induction hypothesis, and the recursion in (28),

$$\begin{aligned}\underline{\beta}_t^{\hat{\pi}^*}(h_t, a_t) &\leq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\leq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^{\hat{\pi}^*}(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^{\hat{\pi}^*}(h_t, a_t),\end{aligned}$$

and then, exactly as in the base case,  $\underline{\alpha}_t^{\hat{\pi}^*}(h_t) \leq \underline{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \leq \tilde{\beta}_t^{\hat{\pi}^*}(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t)$ . Symmetrically, by (24), the induction hypothesis, and the recursion in (29),

$$\begin{aligned}\beta_t^*(h_t, a_t) &\geq \Delta_t(h_t, a_t) + \int \underline{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) \\ &\geq \Delta_t(h_t, a_t) + \int \tilde{\alpha}_{t+1}^*(h_{t+1}) P_{Y,t}(dy_{t+1} \mid h_t, a_t) = \tilde{\beta}_t^*(h_t, a_t),\end{aligned}$$

and then  $\underline{\alpha}_t^*(h_t) \geq \underline{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\beta}_t^*(h_t, \hat{\pi}_t^*(\sigma_t(h_t))) \geq \tilde{\alpha}_t^*(h_t)$ , where the last inequality is the upper sandwich side (31). This establishes (32) for all  $t$ . Combining with the bridge identity of Step 2,

$$V_t^\pi(h_t) - V_t^*(h_t) = \tilde{\alpha}_t^{\hat{\pi}^*}(h_t) - \tilde{\alpha}_t^*(h_t) \geq \underline{\alpha}_t^{\hat{\pi}^*}(h_t) - \underline{\alpha}_t^*(h_t),$$

which is (27).  $\square$